# University physics – my 1978 Sussex B.Sc. revisited and brought up to date

In 1978 I graduated with a B.Sc. (Hons) in physics from the University of Sussex at Brighton. Almost half a century later I have tried to recreate the courses – not as exam notes, but as they would look today: with the history behind them, the central equations, interactive figures you can experiment with yourself, and what has happened in each subject since 1978. Every page can also be downloaded as a PDF.

## Sussex in the 1970s

The University of Sussex was founded in 1961 as one of Britain's new "plate glass" universities, with buildings by Basil Spence in the park at Falmer outside Brighton. Physics was organised in the *School of Mathematical and Physical Sciences*, and teaching was built on the British tutorial system: lectures supplemented by weekly tutorials in small groups, where you handed in problems and had to defend your solutions to a researcher.

> It was Anthony (Tony) Leggett who interviewed me and offered me my place at Sussex, and he was my tutor in elementary particle physics. He received the Nobel Prize in 2003 for his theory of superfluid helium-3 – work he was doing at Sussex in exactly those years – and he died on 8 March 2026. Among my other teachers was Colin Finn. In a way, the pages below are a greeting to them.

The courses are arranged here by subject rather than by year of study. The first twelve pages were written for this collection; the others are earlier pages on the site that cover the rest of a physics degree.

## Classical physics

**Analytical mechanics**

Lagrange, Hamilton and the principle of least action – mechanics rewritten so that it points towards quantum mechanics.

Lagrange · Hamilton · Least action · Newton's Principia

### Fluid mechanics

Bernoulli, Euler and Navier–Stokes, the Reynolds number, Prandtl's boundary layer, lift and circulation, water waves and turbulence.

PDF · Wind tunnel · Towing tank

**Since 1978:** computer simulation (CFD) has taken over much of the wind tunnel's work; whether Navier–Stokes always has smooth solutions became one of the Clay Institute's Millennium Prize Problems in 2000.

### Continuum mechanics

The stress tensor, Mohr's circle, elastic moduli, P and S waves, Euler's buckling load, fracture and viscoelasticity.

PDF · Earthquakes

**Since 1978:** finite-element calculations on a PC have become the engineer's standard tool; auxetic materials with negative Poisson's ratio (1987) and 3D-printed metamaterials.

### Electromagnetism

From Coulomb and Ørsted to Faraday and Maxwell's equations: field lines, electromagnetic waves, the Poynting vector, resonant circuits and fields in matter.

PDF · The spectrum

**Since 1978:** metamaterials with negative refractive index (2000); in the 2019 SI reform the elementary charge became exact – and $\mu_0$ a measured quantity.

## Heat and matter

### Thermodynamics

The laws, cycles and Carnot, entropy, free energies and the Maxwell relations, phase transitions.

PDF

**Since 1978:** the fluctuation theorems for small systems (Jarzynski 1997, Crooks 1999); Landauer's principle on the heat cost of information has been measured experimentally (2012).

### Statistical mechanics

Boltzmann and Gibbs, the partition function, Maxwell–Boltzmann, Bose–Einstein and Fermi–Dirac, the Ising model and critical phenomena.

PDF

**Since 1978:** Wilson's renormalisation group (Nobel 1982); Bose–Einstein condensation in atomic gases (1995); Parisi's theory of spin glasses (Nobel 2021).

### Properties of matter

Forces between molecules, van der Waals and critical points, surface tension, viscosity, elasticity and transport.

PDF · Cornflour and water

**Since 1978:** "soft matter" became a field (de Gennes, Nobel 1991); quasicrystals (Shechtman 1982, Nobel 2011).

### Solid state physics

Crystal lattices and X-ray diffraction, phonons, free electrons, band theory, semiconductors and magnetism.

PDF

**Since 1978:** the quantum Hall effect (1980), high-temperature superconductors (1986), graphene (2004), topological insulators – and the blue LED (Nobel 2014).

### Plasma physics

Debye length and plasma frequency, particle orbits and drifts, magnetohydrodynamics, waves and fusion.

PDF · Aurora

**Since 1978:** JET ran from 1983 to 2023 and set the record of 69 MJ of fusion energy; NIF achieved ignition in 2022; ITER is under construction.

### Superconductivity and superfluidity

Kamerlingh Onnes, Meissner, London, BCS theory, the Josephson effect, helium-4 and helium-3.

PDF

**Since 1978:** Leggett's Nobel Prize (2003) for superfluid helium-3; copper oxides (1986), iron-based superconductors (2008) and hydrides under extreme pressure.

# Quantum physics and light

## Quantum mechanics

From Planck and Bohr to Schrödinger and Heisenberg: wave functions, potential wells, tunnelling, spin, the hydrogen atom, entanglement and Bell's inequality.

PDF

**Since 1978:** Aspect's Bell experiments (1982), the loophole-free Bell test (2015) and the Nobel Prize of 2022; quantum computers and quantum cryptography have become engineering.

## Atomic physics

Spectral lines and Bohr's model, hydrogen orbitals, fine structure and spin, the Zeeman effect, the periodic table, Moseley and atomic clocks.

PDF · Spectra

**Since 1978:** laser cooling and Bose–Einstein condensates (1995), optical atomic clocks with 18-digit precision, trapped ions and Rydberg atoms as quantum computers.

## Optics

Fermat's principle, refraction and the rainbow, matrix optics, interference, N-slit diffraction, Fresnel's equations and resolving power.

PDF · Lenses · Fourier optics

**Since 1978:** adaptive optics, Hubble (1990), super-resolution microscopy (Nobel 2014) and the Event Horizon Telescope's image of a black hole (2019).

## Laser physics

Stimulated emission, population inversion, rate equations, resonators and modes, Gaussian beams and pulses.

PDF · The laser

**Since 1978:** laser cooling (Nobel 1997), frequency combs (2005), optical tweezers and amplified ultrashort pulses (2018), attosecond pulses (2023).

## Nuclei, particles and relativity

### Nuclear physics

The structure of the nucleus, binding energy and the liquid-drop model, the shell model, radioactivity, fission and fusion.

PDF

**Since 1978:** the periodic table is complete up to element 118; the neutron-star merger of 2017 showed where gold is made.

### Particle physics and group theory

Symmetries and groups, SU(2) and SU(3), the quark model, the Standard Model and the Higgs mechanism.

PDF · String theory

**Since 1978:** the W and Z bosons (1983), the top quark (1995), neutrino masses (1998) and the Higgs boson (2012).

### Relativity

Special and general relativity: the Lorentz transformation, spacetime, the equivalence principle, curvature and black holes.

PDF

**Since 1978:** GPS corrects for relativistic effects every day; the accelerating expansion of the universe (1998), gravitational waves (2015) and the image of a black hole (2019).

## What has happened since 1978?

When I took my degree, the Standard Model was just becoming accepted, the charm quark was four years old, and the bottom quark had been discovered the year before. Pocket calculators were new, and “computer” meant a room full of punched cards. The most striking thing on re-reading the courses is not that the foundations have changed – Maxwell, Boltzmann and Schrödinger stand where they stood – but how much of what were thought experiments then is laboratory work today: single atoms trapped with light, entangled photons over hundreds of kilometres, gravitational waves from colliding black holes. At the same time some of the big questions are the same: turbulence, the interpretation of quantum mechanics, the relationship between quantum theory and gravity, and what dark matter and dark energy are. Some even speak of a crisis in physics – see Is physics in crisis?

None of the pages replaces a textbook; each ends with suggestions for further reading. They are meant as an overview for school students thinking of studying physics, for students who want to see how it all fits together, and for old physicists who – like me – want to see where the subject has got to.

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